Sharp subsidies for protecting core maxima in three-sided assignment markets

Maxwell Flanagan
New York City, United States
maxwell.flanagan@gmail.com

17 October 2026

Abstract

Consider three equally sized finite sectors, nonnegative pair values, and productive teams containing exactly one player from each sector. A team's value is the sum of its three pair values; coalition worth is maximum integral matching worth. Assume the original core is nonempty and grand worth is $T > 0$. For each player let H_i be their maximum payoff over that entire original core. We determine the worst-case external subsidy needed to cover every original coalition while paying each player at least $H_i/3$. Every finite market requires strictly less than $T/3$, and the supremum of the subsidy-to-worth ratio is exactly $1/3$. A direct fractional-packing argument combines six core-maximum comparisons with residual budgets at separately attaining core allocations. For an integral partial matching leaving $k \geq 1$ players unmatched in each sector, the associated subsidy obstruction is at most $2(k-1)T/[3(2k-1)]$. This bound is sharp for every $k \geq 2$. We give complete proofs, a family with pair values in $\{0, 1, 2\}$, and reproducible exact checks. A compact supplement records exact nonnegative slack identities and the verification scope. The original core is fixed before subsidy, and the result is static: it makes no claim about reporting incentives across profiles.

Keywords: assignment game; core; individual payoff maximum; subsidy; cost of stability; fractional matching.

1 Introduction

A stable allocation need not simultaneously give each player a substantial fraction of what that player could obtain at another stable allocation. An external budget permits a different question: how much must be added to the grand coalition's worth to protect individual benchmarks while retaining all original coalition constraints?

We study the benchmark $H_i/3$, where H_i is player i's maximum over the entire original core of a three-sided assignment market. The reference core is fixed before intervention. In particular, the benchmark is not recomputed after adding the subsidy. The restriction to nonnegative pair-additive team values is central; an arbitrary three-dimensional value array is not assumed to satisfy our conclusion.

Our main result gives the sharp worst-case subsidy ratio $1/3$. It is a supremum over finite markets of arbitrary size, not a maximum at a finite size. The upper proof applies directly to fractional partial matchings in the subsidy dual. It does not require a reduction to sparse markets, replication of players, or an acyclic support. A second result determines a sharp upper bound for the obstruction supplied by an integral partial matching, indexed by its number of unmatched players per sector.

The proof uses two estimates for the same dual expression. If W is a fractional matching's value, S the sum of residual-weighted individual maxima and β a residual-weighted diagonal pair budget, then

$$3W + S \leq 5T - \beta, \qquad 3W + S \leq 3T + \beta.$$

The first estimate aggregates six comparisons between individual maxima; the second uses a different original-core witness for each maximum. Averaging eliminates β. Retaining the slack terms proves strictness in every finite market and the sharper integral bound.

Scope of this article. The main paper contains the full three-sided theorem, strict finite-market upper bound, integral obstruction refinement and matching family. Online Resource 1 records exact slack identities and reproducibility details. The project's broader October 7 compendium also contains higher-sector, flow, transportation and ancillary core results. Those separate extensions are not included or claimed as additional contributions of this article.

Related work and contribution. The assignment-game connection between core allocations and linear-programming duality is classical (Shapley and Shubik, 1971). Protection relative to individual core maxima is also an established question. Núñez and Rafels (2004, *Bargained stable allocations in assignment markets*, Barcelona Economics Working Paper 153, Definition 6; inspected working version) study the bilateral half-optimal core: stable allocations paying every agent at least half their maximum core payoff. Thus neither the benchmark based on core maxima nor simultaneous fractional protection is introduced here.

External payments to stabilize a game are studied through the cost of stability (Bachrach et al., 2009, arXiv:0907.4385; published successor (Bachrach et al., 2018)). Meir, Rosenschein and Malizia (Meir et al., 2011, Section 2.2) also define the cost of extending a given efficient payoff vector to a stable one; the same definition appears in Bachrach et al. (Bachrach et al., 2018, Section 5). Their input vector has total T; our floor vector $H/3$ may have smaller total. Section 7 records the exact comparison. Bejan and Gómez (Bejan and Gómez, 2009) define minimal grand-coalition expansions and the extended core. The published definitions in Section 2, pp. 5–6, minimize the required expansion over efficient payoff vectors; the extended core coincides with the core for balanced games. These passages were inspected in the author-uploaded published-layout text and agree with the previously read August 1, 2008 author version. Here that ordinary minimum is zero. Our additional restriction is the simultaneous floor $H/3$, which need not itself be efficient or stable. Priority-game transformations also impose individual protections. In the final version of Núñez Lugilde et al. (2024), Definitions 2 and 4 use a prior core allocation; footnote 3 extends the formulas to imputations. Lemmas 2 and 8 identify restrictions of the original core while retaining grand-coalition worth. These statements do not supply the sharp subsidy constant for our possibly incompatible floors. The subsidy framework and the operation of imposing individual claims are established antecedents.

Atay and Núñez (Atay and Núñez, 2019, p. 273) describe a model in Quint (Quint, 1991) in which full teams have pair-additive values and smaller coalitions have zero worth. Their own graph model permits productive partial-sector teams and therefore has different coalition constraints. Results on core extrema in supermodular multisided markets with local monotonicity (Martínez-de-Albéniz et al., 2026) impose additional hypotheses absent here.

The contribution proposed for review is the aggregate compatibility inequality and the resulting sharp subsidy bound for the stated class. The extremal family was obtained earlier in this project's subsidy work; the matching upper bound and its integral refinement complete its extremal interpretation. This manuscript gives a self-contained proof rather than claiming a new general subsidy framework or a new assignment model. A companion manuscript, *Acyclicity and simultaneous*

protection of individual core maxima, asks which fractions of the same benchmarks can be protected *within* the original core and studies graph structure. Its triangle construction differs from the uniform-ideal family used here. The present paper fixes the fraction at one third and optimizes external subsidy. Another companion, *Sharp subsidies for truthful production and network cooperation*, links payments across report profiles; this static result supplies no such incentive guarantee. Historical priority of the precise result remains under review. In particular, full-original comparisons with Quint (Quint, 1991), Lucas (Lucas, 1995) and Stuart (Stuart, 1997) remain incomplete. Other original-source leads and general implication routes are listed in the companion source ledger; no exhaustive priority clearance is claimed.

2 Model and subsidy duality

Let N_1, N_2, N_3 be disjoint sectors of size $n \geq 1$, and put $N = N_1 \cup N_2 \cup N_3$. A market M is specified by nonnegative real matrices A, B, C on $N_1 \times N_2$, $N_1 \times N_3$, and $N_2 \times N_3$, respectively. A full team $e = (i_1, i_2, i_3)$ has value

$$v(e) = A_{i_1 i_2} + B_{i_1 i_3} + C_{i_2 i_3}. \tag{1}$$

For a coalition $Q \subseteq N$, its worth $v(Q)$ is the maximum sum of values of player-disjoint full teams contained in Q. The empty matching is allowed. Pairs and all other coalitions containing no full team have worth zero, even when a constituent pair matrix entry is positive.

Write $T = v(N) > 0$. We assume throughout that the original core

$$\mathcal{C} = \{x \in \mathbb{R}^N : x(N) = T,\ x(Q) \geq v(Q) \text{ for every } Q \subseteq N\}$$

is nonempty, where $x(Q) = \sum_{i \in Q} x_i$. Nonnegativity and full-team coverage describe all coalition constraints:

$$\mathcal{C} = \{x \in \mathbb{R}^N_+ : x(N) = T,\ x(e) \geq v(e) \text{ for every full team } e\}. \tag{2}$$

Indeed, sum the full-team inequalities over an optimal integral matching inside any coalition and use nonnegative payoffs for unmatched players. The reverse implication follows from singleton and full-team coalitions. The core is a compact subset of the nonnegative total-T simplex. Define the attained individual maxima

$$H_i = \max_{x \in \mathcal{C}} x_i. \tag{3}$$

The protected subsidy is

$$s(M) = \min\{y(N) - T : y_i \geq H_i/3 \ (i \in N),\ y(e) \geq v(e) \ (e \text{ a full team})\}. \tag{4}$$

This feasible linear program has a finite minimum. Each feasible vector is nonnegative and covers all original coalitions, including N, so $s(M) \geq 0$. It is interpreted as an allocation for a subsidized grand coalition; it need not belong to the original core.

Let $\mathcal{P}$ be the fractional partial-matching polytope:

$$\mathcal{P} = \left\{ \lambda \geq 0 : p_i := \sum_{e \ni i} \lambda_e \leq 1 \quad (i \in N) \right\}.$$

For $\lambda \in \mathcal{P}$ put $q_i = 1 - p_i$ and define

$$W = \sum_e \lambda_e v(e), \qquad S = \sum_i q_i H_i, \qquad V = 3W + S. \tag{5}$$

Lemma 2.1 (Protected subsidy dual)**.** *For every market satisfying the preceding assumptions,*

$$s(M) = \max_{\lambda \in \mathcal{P}} (W + S/3 - T) = \max_{\lambda \in \mathcal{P}} (V/3 - T). \tag{6}$$

Both extrema are attained.

Proof. Substitute $y_i = H_i/3 + z_i$ in (4). The variable part is to minimize $\sum_i z_i$ subject to $z_i \geq 0$ and $\sum_{i \in e} z_i \geq v(e) - \sum_{i \in e} H_i/3$ for every team e. Its dual has one nonnegative variable λ_e per team and constraints $\sum_{e \ni i} \lambda_e \leq 1$. Adding the constant $\sum_i H_i/3 - T$ to the dual objective gives (6). The primal is feasible and bounded below. Moreover, $\mathcal{P}$ is nonempty and compact, since there are finitely many teams and $0 \leq \lambda_e \leq 1$. Finite-dimensional linear-programming duality proves the assertion, for real as well as rational market data. □

Theorem 2.2 (Sharp subsidy ratio)**.** *Every market in the stated class satisfies $s(M) < T/3$. Nevertheless,*

$$S_3 := \sup_M \frac{s(M)}{v(N)} = \frac{1}{3}. \tag{7}$$

The supremum is approached by markets with pair values in $\{0, 1, 2\}$.

We prove the upper bound in Section 4 and construct the sharpness family in Section 6.

3 Diagonal budgets and comparisons of maxima

Choose an optimal integral matching. Because sector sizes are equal, it can be completed to a perfect matching; nonnegative team values imply that the completion remains optimal. Relabel the players so that this perfect matching consists of $(1, j), (2, j), (3, j)$ for $j = 0, \ldots, n-1$. Let

$$d_j = A_{jj} + B_{jj} + C_{jj}, \qquad \sum_j d_j = T, \tag{8}$$

$$u_{1,j} = A_{jj} + B_{jj}, \qquad u_{2,j} = A_{jj} + C_{jj}, \qquad u_{3,j} = B_{jj} + C_{jj}.$$

Thus $u_{a,j}$ is the diagonal pair budget incident to player (a, j), and

$$\sum_{a,j} u_{a,j} = 2T. \tag{9}$$

Every diagonal team is tight at every $x \in \mathcal{C}$: their covering inequalities sum to the efficiency equality. Consequently

$$\sum_{a=1}^{3} x_{a,j} = d_j, \qquad 0 \leq H_{a,j} \leq d_j, \qquad \delta_{a,j} := d_j - H_{a,j} \geq 0. \tag{10}$$

Zero diagonal values cause no difficulty; no argument divides by d_j.

Lemma 3.1 (Same-sector range)**.** *For every $x \in \mathcal{C}$, sector a, and labels j, h,*

$$x_{a,j} - x_{a,h} \leq u_{a,j}. \tag{11}$$

Proof. Complete the two players of the diagonal team at j outside sector a with player (a, h). By pair nonnegativity this team's value is at least the opposite diagonal pair value, namely $d_j - u_{a,j}$. Core coverage and diagonal tightness give $x_{a,h} + d_j - x_{a,j} \geq d_j - u_{a,j}$. This is (11); it also holds when $j = h$. □

Lemma 3.2 (Six ideal comparisons)**.** *For a full team $e = ((1, j_1), (2, j_2), (3, j_3))$ and every ordered permutation (a, b, c) of $(1, 2, 3)$,*

$$H_{c,j_c} - H_{c,j_b} \geq v(e) - d_{j_b} - u_{a,j_a}. \tag{12}$$

The labels j_1, j_2, j_3 need not be distinct.

Proof. Subtract diagonal tightness at label j_b from coverage of e. For every original core point x this gives

$$x_{c,j_c} - x_{c,j_b} \geq v(e) - d_{j_b} + x_{a,j_b} - x_{a,j_a} \geq v(e) - d_{j_b} - u_{a,j_a},$$

using Lemma 3.1. Choose a core point attaining the *subtracted* maximum H_{c,j_b}. Its other coordinate is at most H_{c,j_c}, proving (12). A different core point may be used for every comparison. No joint attainment is assumed. □

4 Aggregation for arbitrary fractional matchings

Fix $\lambda \in \mathcal{P}$. In addition to (5), write

$$k = n - \sum_e \lambda_e, \qquad \beta = \sum_i q_i u_i, \qquad C_2 = \sum_i q_i^2 u_i. \tag{13}$$

Here and below $u_i = u_{a,j}$ when $i = (a, j)$. Every sector has the same residual mass: $\sum_{i \in N_a} q_i = k \geq 0$. In particular $\sum_i q_i = 3k$. For each diagonal label let

$$n_j = \sum_{a=1}^{3} p_{a,j}, \qquad H_j^{\Sigma} = \sum_{a=1}^{3} H_{a,j}, \qquad E = \sum_{a,j} \left(\sum_{b \neq a} q_{b,j} \right) \delta_{a,j} \geq 0.$$

The symbol n_j is a fractional load on a diagonal label, not the number of labels n; it may be any number in $[0, 3]$.

Lemma 4.1 (Aggregate ideal comparison)**.** *Every fractional partial matching satisfies*

$$V \leq 5T - \beta - \frac{E}{2}. \tag{14}$$

Proof. Sum (12) over the six ordered sector permutations for one team e. Each participating ideal appears twice positively. The negative cross terms are $-\sum_{a=1}^{3} H_{j_a}^{\Sigma} + \sum_{a=1}^{3} H_{a,j_a}$. Each diagonal value and each incident budget occurs twice. Therefore

$$6v(e) \leq 2 \sum_a d_{j_a} + 2 \sum_a u_{a,j_a} + 3 \sum_a H_{a,j_a} - \sum_a H_{j_a}^{\Sigma}.$$

Multiply by λ_e and sum over teams. With $D_p = \sum_{a,j} p_{a,j} d_j$, $U_p = \sum_{a,j} p_{a,j} u_{a,j}$ and $H_p = \sum_{a,j} p_{a,j} H_{a,j}$, this becomes

$$6W \leq 2D_p + 2U_p + 3H_p - \sum_j n_j H_j^{\Sigma}. \tag{15}$$

All sums retain sector multiplicities, including when team labels coincide.

Add $2S$ to (15). For one label j, the coefficient of $H_{a,j}$ is

$$3p_{a,j} + 2q_{a,j} - n_j = 2 + p_{a,j} - n_j = \sum_{b \neq a} q_{b,j}. \tag{16}$$

Replace $H_{a,j}$ by $d_j - \delta_{a,j}$. Summing (16) over a gives $2\sum_a q_{a,j}$, so

$$3H_p + 2S - \sum_j n_j H_j^{\Sigma} = 2\sum_{a,j} q_{a,j} d_j - E.$$

Finally $D_p + \sum_{a,j} q_{a,j} d_j = 3T$ and $U_p = 2T - \beta$. Thus $2V \le 10T - 2\beta - E$, as required. In particular the sign of the deficit term does not depend on integrality of n_j. □

Lemma 4.2 (Weighted residual witness budget)**.** *If $k > 0$, then for each player i*

$$kH_i \le T - W + (k - q_i)u_i. \tag{17}$$

Consequently,

$$V \le 3T + \beta - \frac{C_2}{k}. \tag{18}$$

Proof. For every $x \in \mathcal{C}$, fractional team coverage implies $\sum_i p_i x_i \ge W$. By efficiency,

$$\sum_i q_i x_i \le T - W =: R. \tag{19}$$

In particular $W \le T$. Fix $i \in N_a$ and take a core point with $x_i = H_i$. Lemma 3.1 gives $x_h \ge H_i - u_i$ for every other $h \in N_a$. Since residual weights and all other-sector payoffs are nonnegative,

$$R \ge \sum_{h \in N} q_h x_h \ge q_i H_i + (k - q_i)(H_i - u_i) = kH_i - (k - q_i)u_i.$$

Here $k - q_i = \sum_{h \in N_a \setminus \{i\}} q_h \ge 0$. The lower bound $H_i - u_i$ need not be positive, and q_i may be zero. This proves (17) using just one attaining witness at a time.

Multiply (17) by q_i and sum over players. The resulting scalar inequalities concern the fixed numbers H_i, so different witnesses can be used. Since $\sum_i q_i = 3k$,

$$kS \le 3k(T - W) + k\beta - C_2.$$

Dividing by k proves (18). □

Proof of the upper assertion in Theorem 2.2. For $k > 0$, average (14) and (18):

$$V \le 4T - \frac{E}{4} - \frac{C_2}{2k}. \tag{20}$$

If $C_2 > 0$, this is strictly less than $4T$. If $C_2 = 0$, each nonnegative summand $q_i^2 u_i$ is zero, hence each $q_i u_i$ is zero and $\beta = 0$. Equation (18) then gives $V \le 3T < 4T$. If $k = 0$, every $q_i = 0$ and (19) gives $W \le T$, again implying $V \le 3T < 4T$.

Thus $V < 4T$ at every point of the compact polytope $\mathcal{P}$. The objective V is linear in λ and attains its maximum. Apply the strict inequality at a maximizing point and use Lemma 2.1 to conclude $s(M) < T/3$. □

5 A sharp bound for integral obstructions

For an integral player-disjoint partial matching P, let U be its unmatched players, $k = n - |P|$, and set

$$J(M, P) = v(P) + \frac{1}{3} \sum_{i \in U} H_i - T.$$

The indicator of P is feasible in $\mathcal{P}$, so $J(M, P) \le s(M)$. An upper bound on $J(M, P)$ alone is not an upper bound on the full subsidy.

Theorem 5.1 (Integral residual-count bound)**.** *For $k \ge 1$,*

$$J(M, P) \le \frac{2(k-1)}{3(2k-1)} T. \tag{21}$$

For $k = 0$, $J(M, P) \le 0$. For every integer $k \ge 2$, equality in (21) *occurs in a market with $n = 2k-1$.*

Proof. For the indicator packing, $q_i \in \{0, 1\}$ and $C_2 = \beta$. The two aggregate bounds imply

$$V \le 3T + \frac{k-1}{k}\beta, \qquad V \le 5T - \beta.$$

Weight these inequalities by $k/(2k-1)$ and $(k-1)/(2k-1)$, respectively, and add. This gives $V \le (4 - 1/(2k-1))T$ and hence (21). The argument includes $k = 1$, where the second weight is zero. For $k = 0$, P is perfect and $J(M, P) = v(P) - T \le 0$. Equality for each $k \ge 2$ follows from the family below with $\ell = k - 1$. □

6 The sharpness family

For an integer $\ell \ge 1$, take $n = 2\ell + 1$ and labels $0, \dots, 2\ell$. Initialize A, B, C to the identity matrix. Define special teams

$$e_0 = (0, \ell, 2\ell), \qquad e_s = (s, \ell + s, s - 1) \quad (1 \le s \le \ell - 1). \tag{22}$$

For every $e = (j_1, j_2, j_3)$ in this list set $A_{j_1 j_2} = 2$, $B_{j_1 j_3} = 1$, and $C_{j_2 j_3} = 2$. All labels within a special team are distinct, so these entries are off-diagonal. The special teams are player-disjoint: their sectorwise label sets are $\{0, \dots, \ell - 1\}$, $\{\ell, \dots, 2\ell - 1\}$, and $\{2\ell, 0, \dots, \ell - 2\}$. In particular, special entries do not conflict or overwrite diagonal entries.

Proposition 6.1 (Exact values)**.** *The market in* (22) *has a nonempty original core and*

$$T = 3(2\ell + 1), \qquad H_i = 3 \ (i \in N), \qquad s(M) = 2\ell. \tag{23}$$

Proof. Consider the bipartite incidence graph with one vertex for each label and one vertex for each special team, joined by label membership. Consecutive special teams share one label; nonconsecutive special teams share none, and each has a private second-sector label. Their label sets together contain every label $0, \dots, 2\ell$. Thus the graph is connected and has $(2\ell + 1) + \ell$ vertices and 3ℓ edges, so it is a tree. This also covers $\ell = 1$.

Fix a label r and a sector $a_0 \in \{1, 2, 3\}$. Root this incidence tree at r, assign r color a_0, and traverse outward. When a special team is reached, its parent label is already colored; assign each of its two other labels the sector in which that label occurs in the team. No assignments conflict

because the incidence graph is a tree. Each special team therefore has at least two labels colored by their designated sectors. With $c(j)$ denoting the resulting label color, set

$$x_{a,j} = \mathbf{1}_{\{a=a_0\}} + 2\mathbf{1}_{\{c(j)=a\}}. \tag{24}$$

This vector is nonnegative, has total $3n$, pays each diagonal team exactly 3, and satisfies $x_{a_0,r} = 3$.

We check every full-team inequality. Each team receives a baseline payoff of 1 from the first term in (24). If it contains two positive off-diagonal pair entries, the shared player lies in the same special team for both entries, because special teams are player-disjoint. The team must therefore be that special team. It has value 5 and receives at least $1 + 2 + 2 = 5$. If two positive pair entries are diagonal, all three labels coincide, and the team's value and payoff are both 3.

Every remaining team has at most one positive off-diagonal entry and at most one diagonal entry. Any off-diagonal entry belongs to a special team. At least one of its endpoints has its designated color, since at least two of the three labels in that special team do. Such a team receives at least 3. Its value is at most 3 if the off-diagonal entry is 2, and at most 2 if that entry is 1. A nondiagonal team with no positive off-diagonal entries has value at most 1, covered by the baseline. These cases exhaust all teams.

The diagonal matching has value $3n$ and x covers every matching with total $3n$, so $T = 3n$ and $x \in \mathcal{C}$. Diagonal tightness bounds every original-core coordinate by 3. As r, a_0 were arbitrary, (24) proves $H_i = 3$ for each player.

Every protected allocation pays at least 1 to each player. The ℓ special teams form a partial matching of value 5ℓ, and its $3n - 3\ell$ unmatched players require at least $3n - 3\ell$. Hence the total payment is at least $3n + 2\ell$, proving $s(M) \geq 2\ell$. Conversely start with payoff 1 to every player and add 2 to the first-sector player of each special team. A special team receives 5; every other team has value at most 3 by the preceding classification and receives at least 3. This allocation covers all coalitions, meets every floor, and has total $3n + 2\ell$. Thus $s(M) = 2\ell$. □

The ratios $2\ell/[3(2\ell + 1)]$ tend to $1/3$, proving the remaining assertion of Theorem 2.2. For the special partial matching, $k = \ell + 1$ and $J(M, P) = 2\ell$, which proves the equality assertion of Theorem 5.1.

A nine-player example. For $\ell = 1$, the three sectors each have labels $0, 1, 2$. Start with $A = B = C = I_3$ and set $A_{01} = 2$, $B_{02} = 1$, and $C_{12} = 2$. The proposition gives $T = 9$ and $H_i = 3$ for all nine players, so every protected floor is one. Paying all nine floors costs nine but gives only three to the special team $(0, 1, 2)$, whose value is five. Covering this team and paying the six unmatched players their floors therefore requires at least $5 + 6 = 11$ in total.

Table 1: A minimum-cost protected allocation for $\ell = 1$; columns identify sectors and rows identify player labels.

Label	Sector 1	Sector 2	Sector 3
0	3	1	1
1	1	1	1
2	1	1	1

Table 1 attains this lower bound: the special team receives five, and every other team has value at most three and receives at least three. Thus the subsidy is exactly two, or $2/9$ of original worth. The extra budget covers the conflict between the fixed individual floors and team stability; the

allocation totals eleven and is not a point of the original total-nine core. The benchmarks remain those of that original core.

Remark 6.2 (Why the finite bound is integral). The formula in (21) cannot be applied by inserting a fractional packing's residual mass k. In the family with $\ell = 1$, let D be the diagonal matching and $P = \{e_0\}$. The packing $\lambda = \frac{1}{2}\mathbf{1}_D + \frac{1}{2}\mathbf{1}_P$ has $T = 9$, $W = 7$, and $k = 1$. Its six residual players have $q_i = 1/2$ and $H_i = 3$, so $S = 9$ and $W + S/3 - T = 1$. Substituting $k = 1$ in (21) would incorrectly give zero. The fractional proof instead retains C_2, which need not equal β.

7 Relation to subsidy frameworks and scope

Let $l_i = H_i/3$. From diagonal caps, $\sum_i l_i \le T$. Following Meir et al. (Meir et al., 2011, Section 2.2) and Bachrach et al. (Bachrach et al., 2018, Section 5), let $\mathrm{CoS}(x, G)$ be the least subsidy allowing a stable payoff vector above an efficient preimputation x, with $\sum_i x_i = T$. The following identity is a direct comparison of feasible sets:

$$s(M) = \min_{x \ge l,\ x(N)=T} \mathrm{CoS}(x, G). \tag{25}$$

Indeed, if $y \ge l$ is stable with total at least T, there is a vector x between l and y with total T: distribute $T - l(N)$ of the available nonnegative coordinate increments $y - l$. Conversely, a stable extension of any such x lies above l. In particular the intervention fits an existing subsidy framework. Its sharp quantitative behavior here comes from the structure of the assignment market and its original maxima.

Our universal statement allows arbitrary nonnegative real pair values, dense supports, repeated diagonal labels within a team, and zero diagonal values. It does not assume that all maxima on one sector are jointly attainable. Nonempty original core is an explicit hypothesis, not a claim for all pair-additive three-sided markets. Pair-additivity and nonnegativity enter through the incident-budget identity (9) and the completion argument in Lemma 3.1. Equal sector sizes give the perfect diagonal matching and common residual mass. No extension to signed pair values, arbitrary triple arrays, or a different productive-coalition convention is asserted.

The finite integral bound does not determine the largest full subsidy at each fixed n, because its maximizing dual may be fractional. A constructive allocation formula avoiding general linear programming, and the sharp tradeoff for floors αH_i at other values of α, are separate questions. They are not required for the sharp constant proved here.

Verification and reproducibility. The proofs above are conventional all-parameter arguments. The accompanying standard-library Python package verifies the coefficient identities in (16) and the two slack identities in Online Resource 1, including all 27 ordered label triples over three symbols in the sixfold count (covering every coincidence type). For $\ell = 1, \dots, 12$, it reconstructs each separate ideal-attaining core witness and a matching optimal subsidy allocation from the raw pair matrices. That check covers 3,603,804 full-team inequalities and rejects 12 deliberately damaged allocations. Additional rational packing checks include the fractional counterexample in Remark 6.2. These finite checks corroborate, but do not establish, the universal theorem; no proof-assistant formalization is claimed.

Supplementary material. Online Resource 1 (`ESM_1.pdf`) gives the slack identities, a notation recap, and the scope of the exact tests. Online Resource 2 (`ESM_2.zip`) contains the standalone verification programs, frozen inputs, source hashes, and the fresh execution report. The full October 7 compendium remains a separate research record.

Statements and Declarations

Funding. No funding or support was received for this work.

Competing interests. The author declares no competing interests.

Data and code availability. No empirical dataset is used. Exact mathematical inputs and verification code are included in Online Resource 2. This preparation has not newly deposited or submitted the manuscript or its materials.

AI assistance and responsibility. OpenAI Codex assisted mathematical exploration, proof drafting, verification code and manuscript preparation. Anthropic Claude was used for recorded internal reviews during later preparation. The author personally reviewed the manuscript, proofs, code and results. Internal model reviews and finite checks are not human peer review or formal proof verification.

References

A. Atay and M. Núñez. Multi-sided assignment games on m-partite graphs. *Annals of Operations Research* 279:271–290, 2019. https://doi.org/10.1007/s10479-019-03256-5.

Y. Bachrach, E. Elkind, E. Malizia, R. Meir, D. Pasechnik, J. S. Rosenschein, J. Rothe, and M. Zuckerman. Bounds on the cost of stabilizing a cooperative game. *Journal of Artificial Intelligence Research* 63:987–1023, 2018. https://doi.org/10.1613/jair.1.11270.

C. Bejan and J. C. Gómez. Core extensions for non-balanced TU-games. *International Journal of Game Theory* 38:3–16, 2009. https://doi.org/10.1007/s00182-008-0135-4.

W. F. Lucas. Core theory for multiple-sided assignment games. *Duke Mathematical Journal* 81(1):55–65, 1995. https://doi.org/10.1215/S0012-7094-95-08106-X.

F. J. Martínez-de-Albéniz, C. Rafels, and N. Ybern. Maximum and minimum core payoffs for supermodular multisided assignment markets with local monotonicity. *Discrete Applied Mathematics* 379:605–612, 2026. https://doi.org/10.1016/j.dam.2025.10.006.

R. Meir, J. S. Rosenschein, and E. Malizia. Subsidies, stability, and restricted cooperation in coalitional games. *Proceedings of IJCAI 2011*, pp. 301–306. Author-hosted extended manuscript, Section 2.2.

I. Núñez Lugilde, A. Estévez-Fernández, and E. Sánchez-Rodríguez. Priority coalitional games and claims problems. *Mathematical Methods of Operations Research* 100:669–701, 2024. https://doi.org/10.1007/s00186-024-00876-x. The source comparison uses the final Version of Record, Definitions 2 and 4 and Lemmas 2 and 8, pp. 675–676 and 679–680.

T. Quint. The core of an m-sided assignment game. *Games and Economic Behavior* 3(4):487–503, 1991. https://doi.org/10.1016/0899-8256(91)90017-9.

L. S. Shapley and M. Shubik. The assignment game I: The core. *International Journal of Game Theory* 1:111–130, 1971. https://doi.org/10.1007/BF01753437.

H. W. Stuart, Jr. The supplier–firm–buyer game and its m-sided generalization. *Mathematical Social Sciences* 34(1):21–27, 1997. https://doi.org/10.1016/S0165-4896(96)00830-X.